

Prepared for: FamilySearch
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Course: ENSIGN COLLEGE | IT370
Date: June 17, 2026

Technical Infrastructure Proposal:

Heritage Vault 2.0

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Executive Summary

FamilySearch is modernizing its core record-preservation infrastructure by launching the Heritage Vault 2.0 project. This enterprise technical proposal converts critical business constraints into highly secure, fault-tolerant, and high-performance Linux-based system architectures. Because the assets managed by FamilySearch comprise irreplaceable global genealogical records and sacred ordinance data for members of The Church of Jesus Christ of Latter-day Saints (LDS), the underlying systems demand absolute runtime availability, rigorous access control, cryptographic privacy enforcement, massive distributed storage redundancy, and rapid search optimization. This document establishes a production-ready architectural blueprint designed to fulfill these operational objectives.

Problem Statement

FamilySearch operates an expansive digital repository that preserves billions of vital historical records, ancestral family trees, and high-resolution documents. This global platform serves as a critical spiritual utility for LDS Church members conducting ancestral ordinance research, as well as an essential professional archive for genealogical researchers worldwide.

However, the legacy infrastructure experiences severe performance limitations that directly threaten its ongoing mission. First, massive simultaneous spikes in regional traffic place heavy strain on application availability, leading to high response latencies or localized service disruptions during peak usage periods. Second, an increasingly sophisticated external threat landscape exposes sensitive, personally identifiable information (PII) to potential unauthorized access or data exfiltration risks. This necessitates strict system-level alignment with evolving global compliance standards, notably the General Data Protection Regulation (GDPR, 2016) and the California Consumer Privacy Act (CCPA, 2018). Third, the relentless accumulation of high-resolution digital media creates immense storage overhead, intensifying the

baseline risks of data corruption, bit rot, and mechanical disk failures if structural redundancy measures remain inadequate.

Without a modern, multi-layered structural overhaul, FamilySearch faces risks of localized system downtime, record corruption, and regulatory non-compliance, jeopardizing the global availability of sacred historical assets.

User Demographics & High-Volume Access Strategy

The primary user base of Heritage Vault 2.0 includes LDS Church members completing ordinance research alongside international genealogy professionals and hobbyists. This diverse user base creates a highly transactional environment characterized by intensive, concurrent read/write operations executing across multiple geographic regions.

High Availability and Load Balancing Architecture

- **Global Traffic Management (GTM):** Geo-DNS routing (implemented via AWS Route 53 or global traffic managers) dynamically routes incoming global traffic to the nearest regional data center or cloud availability zone based on latency metrics, balancing user traffic evenly across infrastructures.
- **Layer 4/7 Load Balancing:** In each regional data center, a cluster of high-performance Linux instances running HAProxy paired with Keepalived manages the ingress layer. Keepalived employs the Virtual Router Redundancy Protocol (VRRP) to manage a shared Virtual IP (VIP), ensuring instantaneous failover if a primary load balancer experiences a hardware fault (Hinden, 2010). HAProxy terminates incoming SSL/TLS sessions and distributes incoming HTTP/HTTPS traffic across the web application tier utilizing a least-connections algorithm.

- **Application Server Clustering:** The application layer runs on lightweight, containerized Enterprise Linux instances managed via an orchestration platform like Kubernetes or Podman clusters. If an underlying application node suffers a kernel panic or hardware failure, the active load balancer automatically detaches the unhealthy instance within milliseconds, maintaining an uninterrupted user session.

System Security & Privacy Enforcement

Because Heritage Vault 2.0 hosts sensitive, personally identifiable genealogical profiles and sacred historical data, protecting data privacy and ensuring total compliance with global privacy regulations like GDPR and CCPA is a core priority (GDPR, 2016; CCPA, 2018).

- **Mandatory Access Control (MAC):** All enterprise Linux node installations must run SELinux (Security-Enhanced Linux) strictly in Enforcing mode. As detailed in the Ascend Training Series: Linux+ textbook, SELinux enforces fine-grained type-bounded policies that isolate critical system daemons (such as Apache, Nginx, or PostgreSQL) (Red Hat, Inc., 2023). This ensures that even if an application-layer vulnerability is compromised, the attacker is strictly confined and prevented from reading unapproved directory pathways or executing unauthorized system binaries.
- **Host-Based Network Firewalls:** Host firewalls managed via firewalld or nftables will implement a default-drop policy, explicitly filtering all inbound traffic and allowing access only on mandated service ports (e.g., port 443 for HTTPS, and port 22 restricted exclusively to secure bastion host IP ranges). This configuration is reinforced by Fail2ban to automatically drop connections displaying brute-force patterns.

Cryptographic Protections

- **Data-in-Transit:** All client connections require TLS 1.3 encryption leveraging strong cipher suites (e.g., ECDHE-RSA-AES256-GCM-SHA384), conforming to internet security standards outlined in RFC 8446 (Rescorla, 2018).
- **Data-at-Rest:** Storage volumes hosting sensitive records must use LUKS (Linux Unified Key Setup) block-level encryption, integrated with hardware Security Modules (HSMs) for secure, automated key management and rotation.

Administrative Authentication and Access Control

Remote infrastructure management requires mandatory Multi-Factor Authentication (MFA) executed via cryptographic SSH keys combined with time-based one-time password (TOTP) modules via `pam_google_authenticator`. Elevated administrative privileges are strictly constrained within the `/etc/sudoers` configuration utilizing Role-Based Access Control (RBAC) rules to enforce the principle of least privilege.

Data Storage & Redundancy Strategies

Preserving billions of digital historical assets against physical device failures, file-system corruption, and catastrophic regional disasters requires a multi-layered storage architecture.

- **Local Storage Fault Tolerance (RAID):** Critical database index positions and caching tiers will utilize hardware RAID 10 configurations built with enterprise-grade NVMe storage drives. RAID 10 offers an ideal balance of fast random read/write performance alongside excellent fault tolerance, remaining fully operational during multiple simultaneous disk failures as long as the faults occur across different mirrored pairs.

- **Distributed File Storage Tier:** For the massive volume of uncompressed historical document images, a distributed file framework such as Ceph Storage or enterprise-grade hybrid cloud object storage will be utilized. Ceph eliminates the single-point-of-failure vulnerabilities of traditional NAS architectures by leveraging a dynamic CRUSH placement algorithm to distribute data blocks across independent, commodity hardware storage nodes.

Backup, Snapshotting, and Disaster Recovery (DR)

- **Filesystem Snapshots:** Heritage Vault 2.0 leverages advanced file systems (such as ZFS or Btrfs) to execute near-instantaneous hourly, atomic snapshots. This provides rapid recovery pathways against accidental file updates or data corruption without impacting system performance.
- **Offsite Backup Replication:** Automated scripts compress and encrypt daily block-level differential backups, transferring them to isolated, offsite read-only storage repositories. These backup systems use strict immutable policies to provide reliable recovery options against ransomware risks.

Performance Optimization & Scalability

Promptly serving data from massive datasets comprising billions of records requires system-level optimization and multi-tiered caching frameworks.

Linux Kernel and Network Performance Tuning

The standard Linux networking stack will be optimized to handle intensive internet workloads by applying explicit overrides within `/etc/sysctl.conf`:

- **net.core.somaxconn = 4096:** Increases the maximum socket listen queue backlog to gracefully handle unexpected spikes in concurrent user traffic.
- **net.ipv4.tcp_max_syn_backlog = 8192:** Hardens the network engine against SYN flood attempts while maintaining transaction stability.
- **vm.swappiness = 10:** Adjusts virtual memory management behavior to prioritize physical RAM allocations over disk swapping, minimizing storage-induced input/output latency.

Multi-Tiered Caching Infrastructure

- **Edge Caching Tier:** Global Content Delivery Networks (CDNs) will cache static web dependencies and public document media at edge positions closest to users, absorbing up to 80% of routine read traffic.
- **In-Memory Caching Tier:** An active Redis Enterprise cache cluster sits directly in front of the relational storage engine. Redis handles frequently requested family lineages, transient user sessions, and common search results directly in memory, bringing average request responses down to sub-millisecond ranges.
- **Database Scaling Tier:** Relational database systems (such as PostgreSQL clusters) utilize an aggressive horizontal read-scaling strategy. This design routes write operations directly to a primary instance while distributing complex queries across a pool of read-only replica databases.

Technical Implementation Roadmap

To safely introduce these infrastructure updates without disrupting FamilySearch's live production environments, a phased deployment plan is established:

Table 1: Engineering Implementation Phases

Phase	Title	Core Objectives
Phase 1	Platform Hardening & Base OS Layer	Provision standard enterprise Linux images configured with strict SELinux rules, firewalld policies, and LUKS encryption.
Phase 2	Storage Fabrication	Establish local RAID 10 configurations, build out the distributed Ceph storage tier, and verify snapshot and backup automation.
Phase 3	Cluster Deployment & Caching Setup	Install Redis cache clusters and HAProxy instances; distribute containerized workloads across multi-region Kubernetes groups.
Phase 4	Source Control & Environment Isolation	Establish secure Git workflows, isolated development servers, and automated staging test environments to protect production baselines.
Phase 5	Migration, Validation, & Auditing	Perform secure data synchronization, complete comprehensive load and penetration testing, and securely migrate DNS records.

Incorporation of Feedback from Peer Review and Instructor

This finalized proposal incorporates valuable technical recommendations received during the Week 6 review process from peers Antonio Medina and Brodie Busby, as well as the course instructor. These refinements ensure a comprehensive, production-ready blueprint.

Table 2: Adjustments From Draft to Final Report

Source of Feedback	Initial Draft Baseline	Final Report Correction & Improvement
Peer Review (Antonio Medina)	Listed the California Consumer Privacy Act (CCPA) generally alongside global regulations.	Expanded the security and regulatory compliance sections to explicitly define CCPA as a state statute that establishes a legal baseline for how user information must be stored, handled, and deleted.
Peer Review (Antonio Medina)	The deployment plan proceeded directly from storage setup to multi-tier	Added Phase 4 to the Technical Implementation Roadmap, outlining how developers can safely begin

Source of Feedback	Initial Draft Baseline	Final Report Correction & Improvement
	cluster configuration.	building application features using isolated development servers and secure Git source control before production deployment.
Peer Review (Brodie Busby)	Included repeated, identical inline citations for Red Hat publications within the final conclusion summary.	Completed a thorough citation audit across the document to clean up repetitive references, ensuring clean, precise APA 7th Edition styling throughout.
Instructor Feedback (Silas Wyman)	Focused primarily on system software configurations and baseline technical specifications.	Integrated detailed infrastructure cost considerations, hardware redundancy strategies, and resource management models to make the architecture comprehensive for business stakeholders.

Reflection on FamilySearch and the Sacred Nature of the Work

Beyond its technical specifications, the Heritage Vault 2.0 architecture was designed with a deep appreciation for the profound mission of FamilySearch (FamilySearch, 2026). Preserving genealogical records is not merely a data archiving task; it is a sacred effort to gather God’s children across the veil, ensuring that individuals are remembered and linked to their families for eternity.

This perspective directly guided the engineering decisions within this proposal. The emphasis on high availability, multi-region load balancing, and rapid database synchronization is intended to provide a reliable, frictionless experience for patrons worldwide as they search for their ancestors. Similarly, the choice of enterprise-grade NVMe RAID 10 arrays and Ceph distributed storage systems reflects a commitment to protecting these irreplaceable records against data loss or corruption.

Furthermore, the implementation of rigorous security controls—such as SELinux enforcement, LUKS encryption, and TLS 1.3 protections—was approached as an ethical obligation to safeguard the private data of millions of families. By treating this technical framework as a custodian of sacred family histories, the solutions were designed to ensure that the Heritage Vault 2.0 platform remains a secure, stable, and enduring resource for generations to come.

Artificial Intelligence Use & Learning Integration

In alignment with academic guidelines, this section outlines the structured and collaborative role that Artificial Intelligence (AI) tools played in developing this technical proposal.

Prompt Strategy and Information Discovery

AI was utilized as an interactive engineering reference assistant to explore system configurations and optimize Linux configurations. Specific prompt strategies included:

- "Formulate a highly available Linux load balancing design pattern using HAProxy and Keepalived that explicitly handles high read traffic environments."
- "What are the industry-standard Linux kernel configuration variables inside sysctl.conf to optimize a high-volume, transactional web application?"

Implementation Rationales and Verification

AI-generated suggestions were not accepted implicitly; instead, they served as foundational technical benchmarks. Every automated recommendation—such as overriding network socket behaviors via `net.core.somaxconn` or enforcing mandatory access controls with SELinux—was rigorously evaluated against official Linux documentation and the Ascend Training Series: Linux+ textbook (Red Hat, Inc., 2023). This process ensured that all proposed solutions directly addressed the reliability, performance, and security needs of FamilySearch's Heritage Vault 2.0 infrastructure.

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